

Digital Electronic Systems Design Project Report

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Timing Analysis

The implemented design demonstrates the timing analysis yields compliant results for both setup and hold constraints at the target operating frequency of 151 MHz.

Regarding the setup time, the most critical path is located within the `reverb_0` sub-block (Path 21). While the design requirement demands a 151 MHz clock, this implementation achieves a positive slack of 0.299 ns. This safety margin allows the circuit to potentially operate at a maximum frequency of approximately 158 MHz.

Concerning the hold time, the critical path is located within the `axis_fifo_0` (Path 31). The required hold time constraint is 0 ns, and the implementation guarantees a safe positive hold slack of 0.024 ns, ensuring that no race conditions occur.

The detailed parameters resulting from the implementation are summarized in Table 1 for setup time and in Table 2 for hold time below, related to the above mentioned most critical paths.

Table 1: Timing Summary: Setup Time (Path 21)

Parameter	Value	Unit
Slack	0.299	ns
Requirement	6.627	ns
Data Path Delay	2.378	ns
<i>Logic Delay</i>	0.742	ns
<i>Route Delay</i>	1.636	ns
Clock Path Skew	0.055	ns
Clock Uncertainty	0.079	ns

Table 2: Timing Summary: Hold Time (Path 31)

Parameter	Value	Unit
Slack	0.024	ns
Requirement	0.000	ns
Data Path Delay	0.347	ns
<i>Logic Delay</i>	0.141	ns
<i>Route Delay</i>	0.206	ns
Clock Path Skew	0.013	ns

Power Analysis

At the operating frequency of 151 MHz, the following implementation's power consumption is as shown in the following tables.

Table 3: Power Analysis Summary.

Parameter	Value	Unit
Total On-Chip Power	0.334	W
Design Power Budget	N/A	–
Power Budget Margin	N/A	–
Junction Temperature	26.7	°C
Thermal Margin	58.3	°C
Effective θ_{JA}	5.0	°C/W
Power to Off-Chip Devices	0	W
Confidence Level	Low	–

Table 4: On-Chip Power Breakdown.

Component	Power	Unit	Ratio (%)
Dynamic	0.261	W	78%
<i>MMCM</i>	0.126	W	48%
<i>BRAM</i>	0.078	W	30%
<i>I/O</i>	0.017	W	6%
<i>Clocks</i>	0.018	W	7%
<i>Signals</i>	0.012	W	5%
<i>Logic</i>	0.009	W	3%
<i>DSP</i>	0.002	W	1%
Device Static	0.074	W	22%